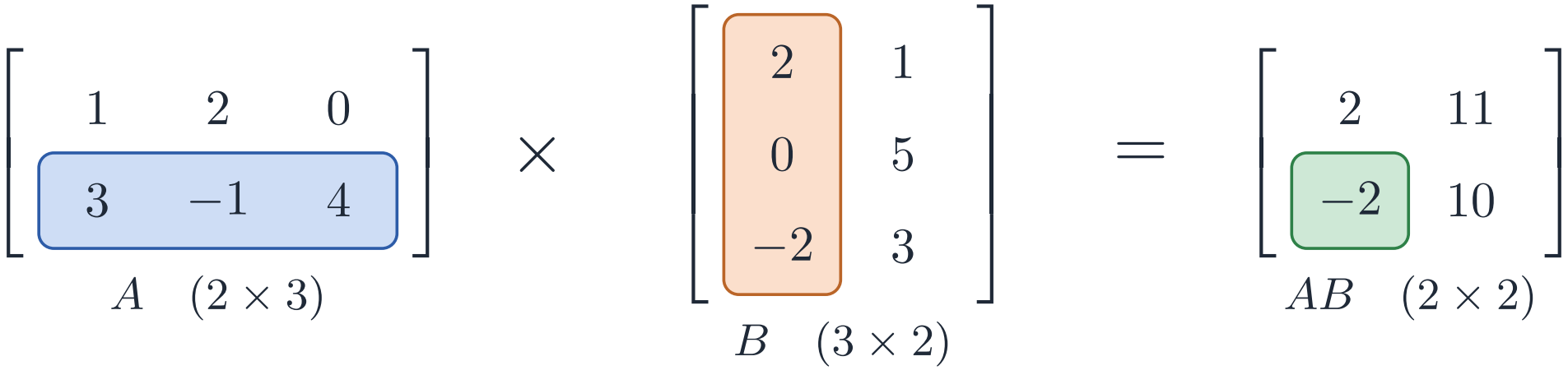


$$c_{21} = \sum_k a_{2k}\, b_{k1} = (3)(2) + (-1)(0) + (4)(-2) = -2$$